3 **(a)** State whether trichloroethanoic acid or ethanoic acid, is the stronger acid. Explain your answer.

.....

.....

.....

.....

.....

..... [2]

(b) Phenyl ethanoate is often used as a solvent and as the building block for the synthesis of other chemicals.

(i) One method of its production involves phenol with an appropriate acid chloride via a **two**-step process. Identify the reactant required for each step.

step 1: .....

step 2: .....
[2]

(ii) Give two reasons why phenol does **not** react with carboxylic acids to form esters.

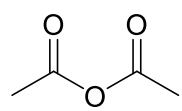
.....

.....

.....

.....[2]

Another method of production involves phenols reacting with a class of compounds known as acid anhydrides. Ethanoic anhydride, $(CH_3CO)_2O$, is an example of a common commercially available acid anhydride.



ethanoic anhydride

Acid anhydrides undergo similar reactions to acid chlorides, and are easier and safer to handle in organic synthesis.

(c) **(i)** Write a balanced chemical equation for the formation of phenyl ethanoate using phenol and ethanoic anhydride.

[1]

(ii) Suggest why acid anhydrides are generally less reactive than acid chlorides towards nucleophiles.

.....

.....

..... [1]

(d) **R** is formed when an acid anhydride **S**, $C_{14}H_{10}O_3$, reacts with CH_3NH_2.

R is a neutral compound and has the molecular formula C_8H_9NO. **R** reacts with hot H_2SO_4(aq) to produce **P** and CH_3NH_2.

P is not very soluble in water, but dissolves after reacting with an excess of Na_2CO_3(aq).

(i) Name the type of reaction occurring when **R** reacts with hot aqueous sulfuric acid.

..... [1]

(ii) **P** has the molecular formula $C_7H_6O_2$. **P** can be formed when methylbenzene reacts with acidified potassium manganate(VII).

Draw the structure of **P**.

[1]

(iii) Write the equation for the reaction of **P** with an excess of Na_2CO_3(aq).

..... [1]

(iv) Suggest the structures of **R** and **S**.

R	**S**

[2]

(v) Use of the *Data Booklet* is relevant to this question.
Infra-red absorptions are useful in identifying functional groups present in molecules.

Suggest an absorption frequency range which can be used to distinguish between molecule **R** and **P**.

..... [1]

[Total:14]